

Marine communities

SIO 30 The Oceans

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Deep-sea ecologist

- BSc in Oceanography
University of São Paulo, SP-Brazil
- MSc in Oceanography
Biological Oceanography curricular group
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- PhD in Oceanography with an Interdisciplinary
Specialization in Environmental Research
Biological Oceanography curricular group,
specialization in Economics
SIO-UCSD

Main interests: Deep sea, Deep-sea ecology & biology, Chemosynthetic ecosystems,
Deep-sea ecosystems connectivity, Trophic ecology, Benthos, Invertebrates, Crustaceans

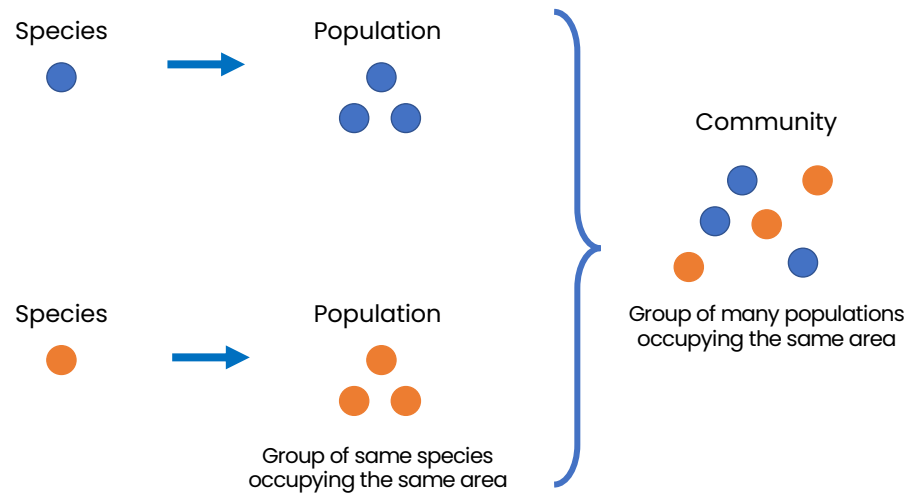
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Marine communities

- Marine community organization
- Shoreline communities
- Shallow-water communities
- Deep-sea communities
- Organisms in marine communities

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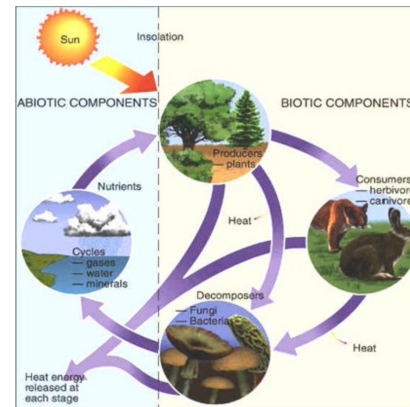
What is a community?



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Physical and biological interactions in a community

- Habitat: physical location, the living space
- Physical factors that affect the environment:
 - Temperature, depth, salinity, light exposure, etc.
- Biological factors that affect the environment:
 - Interactions between organisms (competition, predation, parasitism, commensalism, symbiosis, etc.)
- Niche: occupation within a habitat
 - Biological and physical associations
 - Use of resources, food web position
 - Function or activity



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Range of tolerance

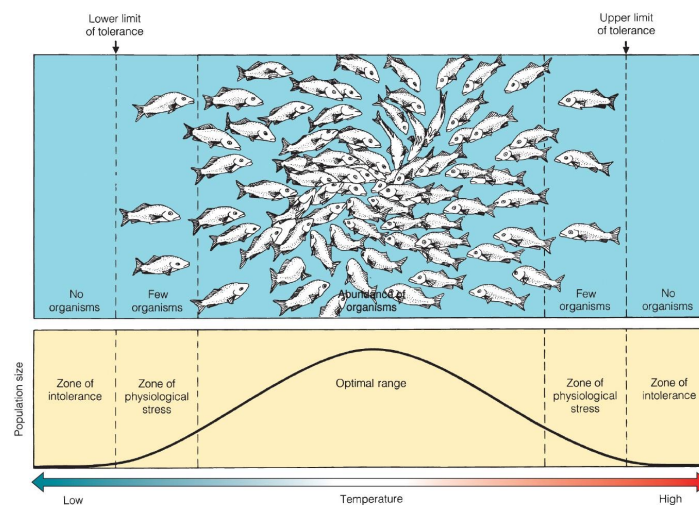
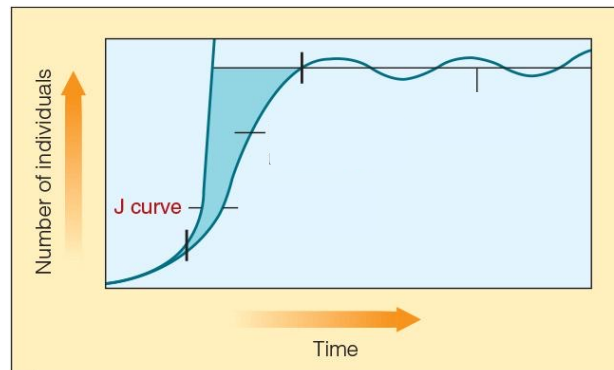


Figure 16.2 Range of tolerance to a physical factor—in this case, temperature—for a population of organisms.

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Growth rate and Carrying capacity

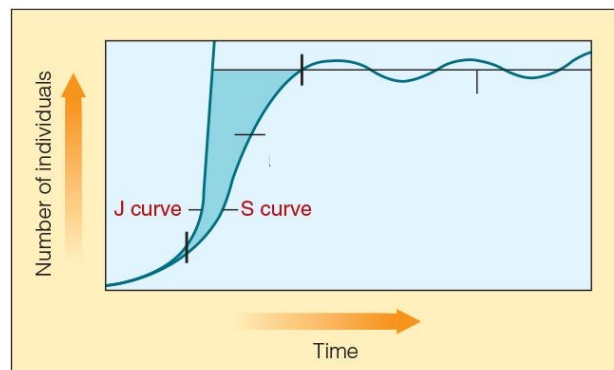
- J curve: population growth without restraints.



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Growth rate and Carrying capacity

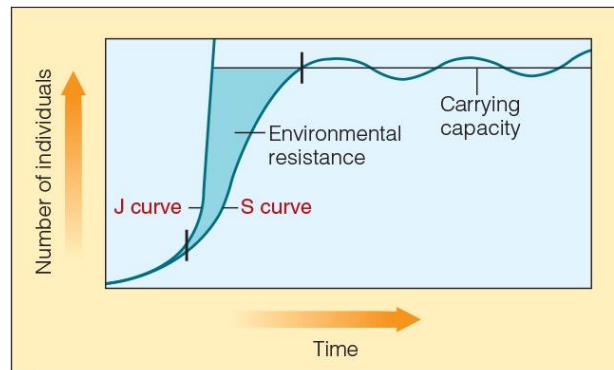
- J curve: population growth without restraints.
- S curve: population growth when there is environmental resistance.



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Growth rate and Carrying capacity

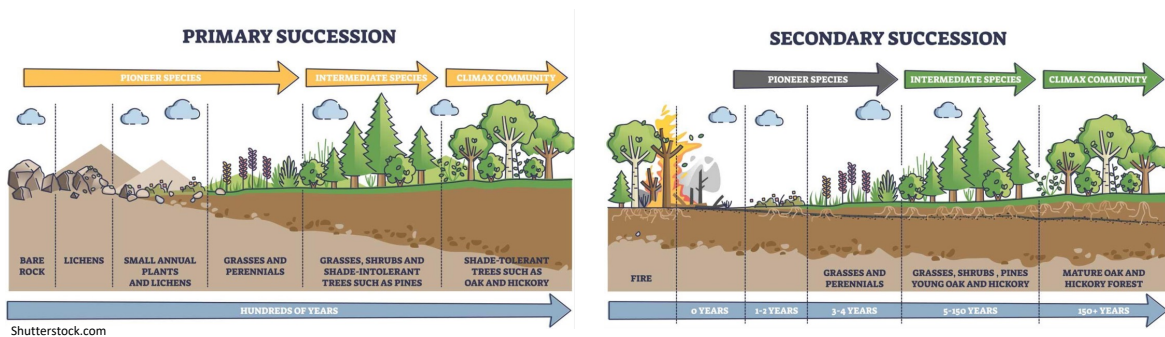
- J curve: population growth without restraints.
- S curve: population growth when there is environmental resistance.
- Carrying capacity: the number of individuals that the environment can support without degradation.



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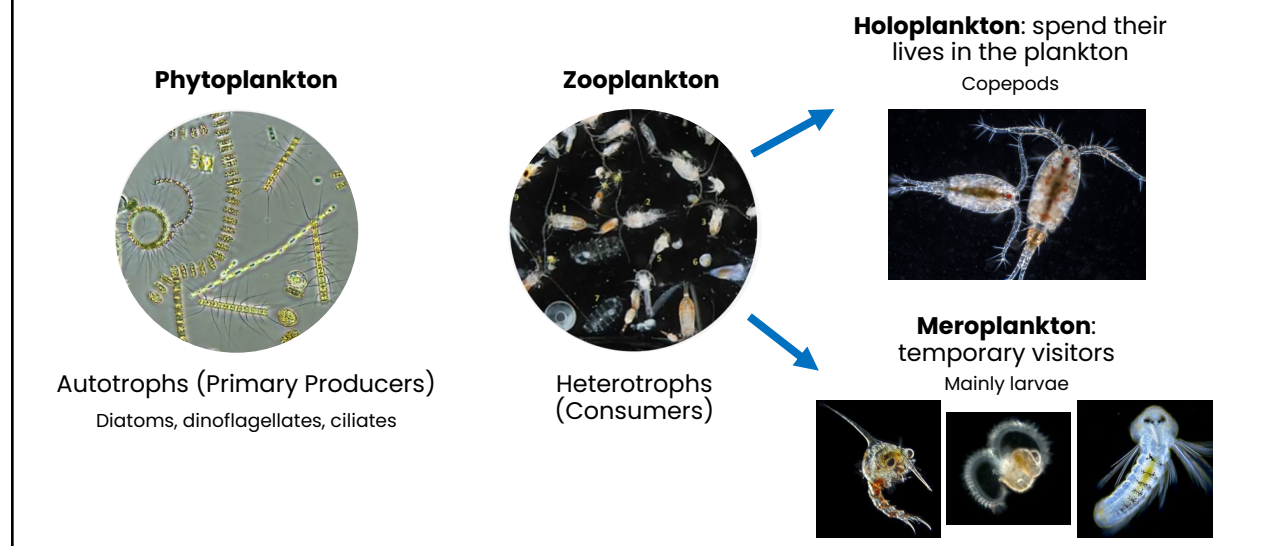
Communities change with time

- Climax communities: long-established, stable community
- Succession: orderly changes in time



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Planktonic communities



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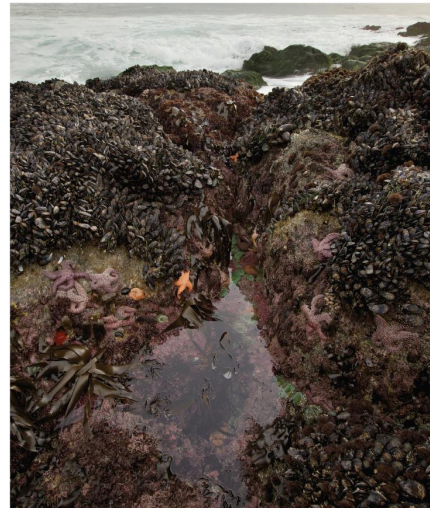
Shoreline communities

Rocky shore, Intertidal zone, Wetlands

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Physical factors affecting shoreline communities

- Wave action
- Exposure to the sun
- Substrate (sandy vs. rocky)
- Desiccation
- Salinity variation
- Temperature
- Oxygen levels



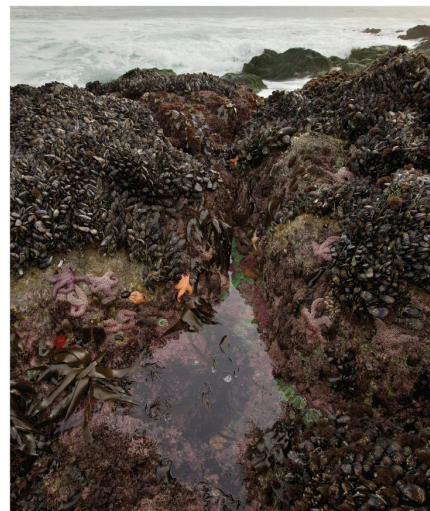
DAVID LUTSCHWAGER/National Geographic Image Collection

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Physical factors affecting shoreline communities

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↑ or ↓ Diversity/
Density?



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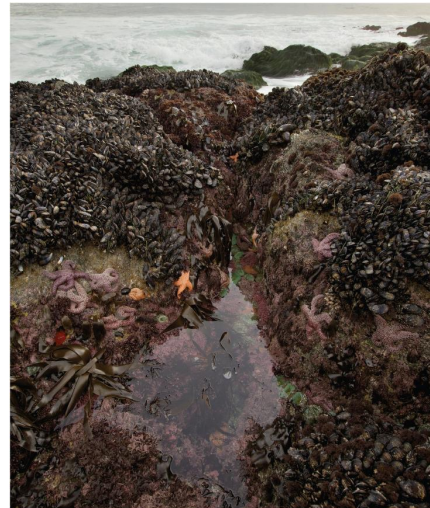
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↑ Diversity/Density

Highly productive, adaptations for sun exposure and salinity & temperature variation

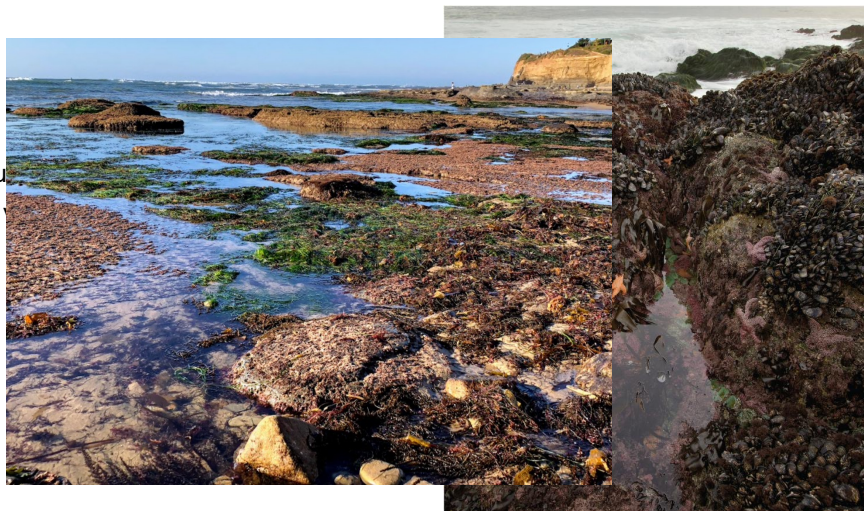


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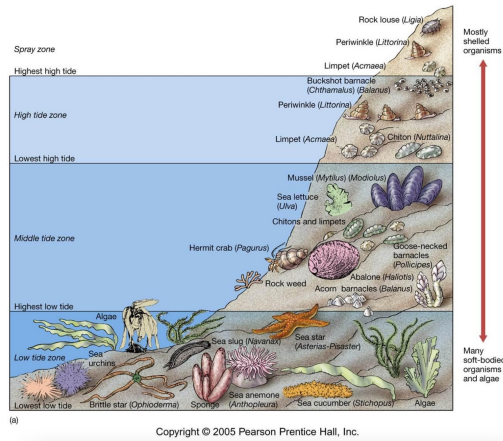


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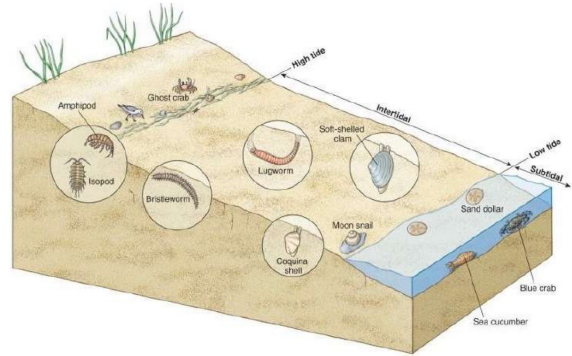
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Intertidal zonation

Rocky shores



Sandy beach



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Wetlands

Estuaries



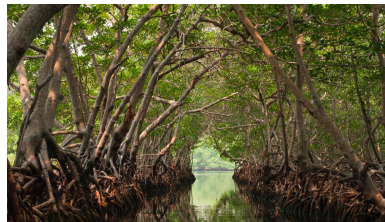
Salt marshes



Sea grasses



Mangroves



- Highly productive
- Nursery habitat
- Great diversity
- Adaptations for sun exposure and salinity variation

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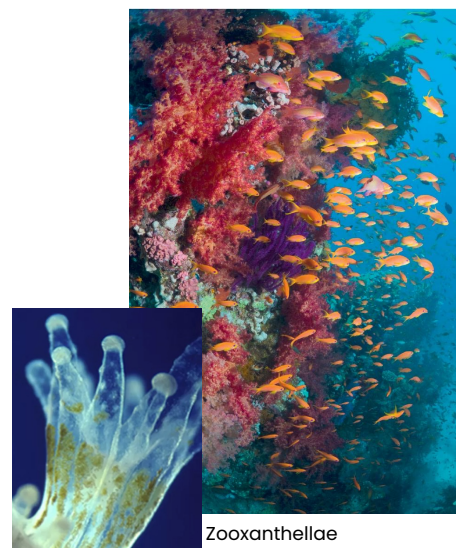
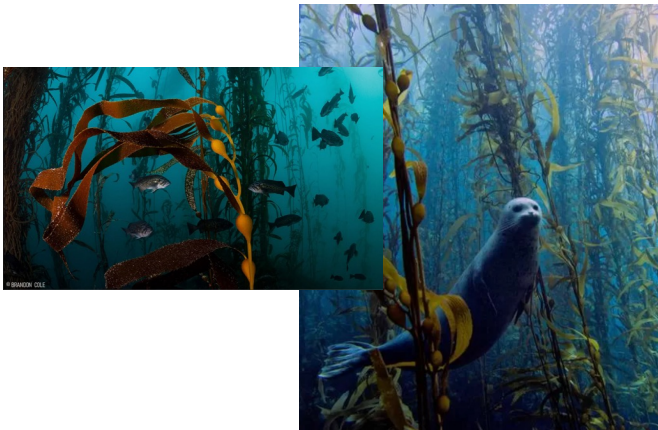
Shallow-water benthic and open-ocean communities

Kelp forests, Coral reefs, Open ocean (pelagic)

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Kelp forests and coral reefs

- Highly productive ecosystems with very dense and diverse community



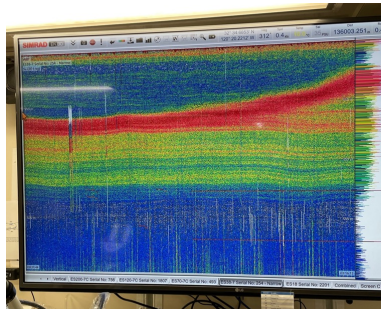
Zooxanthellae

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Open ocean (pelagic)

- 83% of total ocean biomass is concentrated in top 200 meters
- Deep scattering layer: dense layer of animals (fish, shrimps, squids) that migrate up and down

Diel vertical migration



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The Deep Sea

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The deep sea

- > 200 meters deep
- Extreme temperatures (cold and hot)
- Extreme pressure
- No primary production – except chemosynthesis
- No light – except bioluminescence
- Weird animals

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Adaptations to the deep sea

- No eyes or large eyes
- Bioluminescence
- Countershading
- Red color, transparent, or ultra black
- Symbiotic relationships with microbes
- Slow metabolism and growth rates
- Longevity
- **Gigantism**
- Opportunism

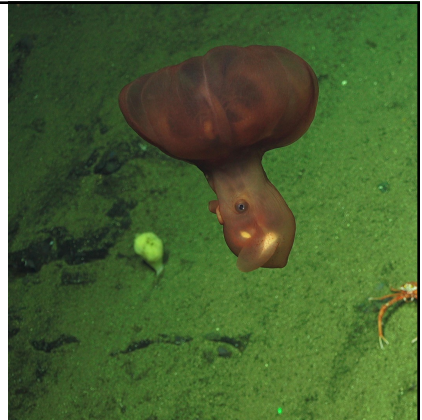
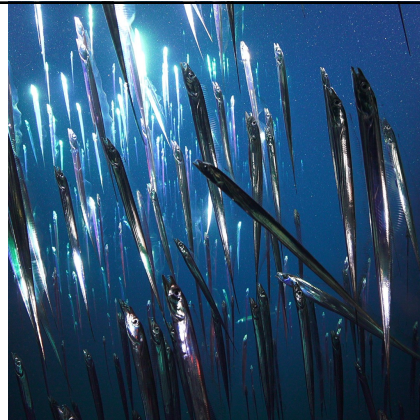


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Deep-sea communities

Deep pelagic, Abyssal plain, Trenches, Seamounts and
Cold-water corals, Chemosynthetic ecosystems

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Deep pelagic

- Fairly uniform and dark
- Increasing pressure with depth
- Limited food (marine snow)
- Biodiversity is relatively low and decreases with depth
- Strategies to blend into the darkness & Bioluminescence
- Connects the deep- and shallow-water ecosystems.
- Threats: deep-sea fishing and mining.

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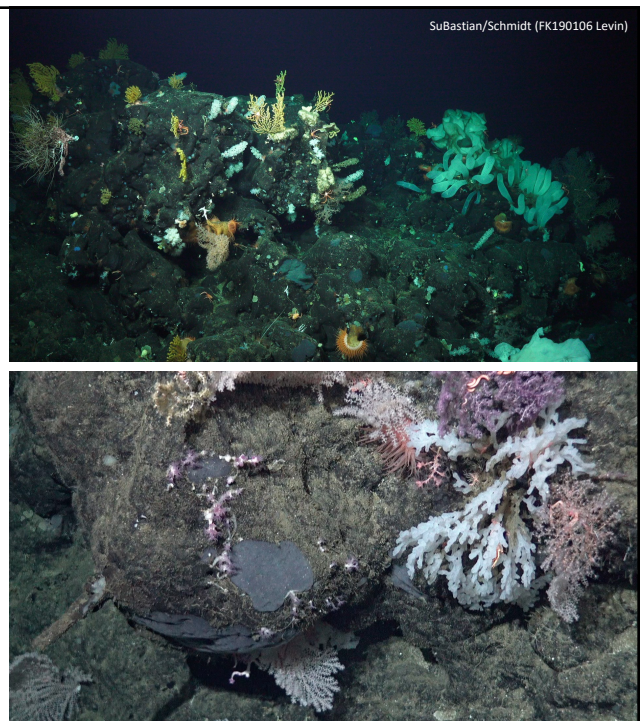
Abyssal plain

- Most of the deep ocean
- Sandy/muddy bottom
- Marine snow is the main source of food
- Not a lot of life compared to other deep-sea and shallow-water ecosystems
- Threats: deep-sea mining

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Seamounts and Cold-water corals

- Found near boundaries of tectonic plates and mid-plate near hotspots
- Hard, rocky bottom
- High nutrient, highly productive -> high biomass
- Economically important fisheries
- Cold-water corals
- Threats: deep-sea fishing, deep-sea mining, coral extraction



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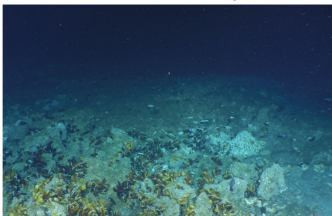
Trenches

- Deepest points on Earth
- Long, narrow steep-sided depressions in the ocean
- Subduction of plates
- Deepest trench: Mariana Trench at 11,034 m
- Longest trench: Peru-Chile Trench (~5,900 km long)
- Pressure 1,000x greater than the surface
- Freezing temperatures
- Accumulation of food

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Chemosynthetic ecosystems

Methane seeps

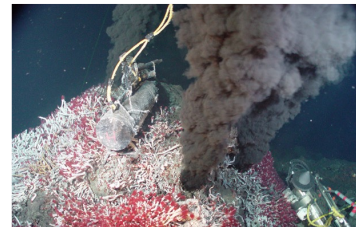


Oxygen Minimum Zones



Gallo & Levin (2016)

Hydrothermal vents



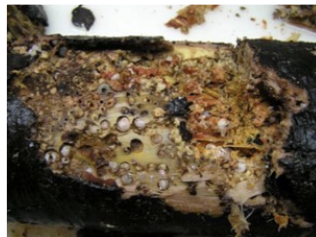
NOAA Ocean Explorer

Whale falls

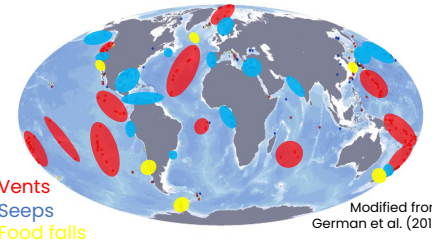


Sumida et al. (2016)

Wood falls



Bienhold et al. (2013)



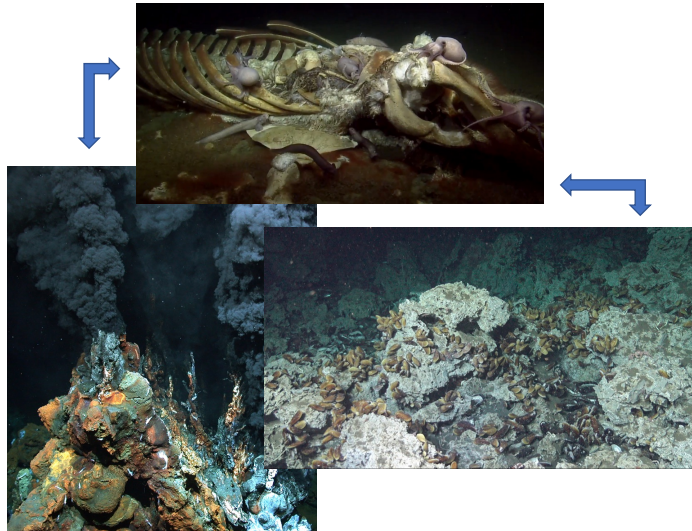
Vents
Seeps
Food falls

Modified from German et al. (2011)

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Connectivity among chemosynthetic ecosystems

- **Chemosynthesis:** conversion of carbon into organic matter using inorganic molecules (e.g., hydrogen, hydrogen sulfide) or methane as an energy source by microbial activity.
- Toxic environment ($\uparrow$ sulfide)
- Specialized, chemosynthetic communities.
- Sporadic/temporary ecosystems.
- Shared fauna at the genus level.
- Stepping-stone hypothesis



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Hydrothermal vents

- Along tectonic plate boundaries (spreading and back arc basins).
- Elevated temperatures ($>400^{\circ}\text{C}$) and sulfide concentration
- Microbial mats, vesicomyid clams, giant tubeworms, Alvinocaridid shrimps, yeti crabs.
- Precipitation of minerals and rare earth elements.
- Threats: deep-sea mining.

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Organic falls

- Fresh supply of food to the deep ocean
- Whale falls are common along whale migration routes
 - Successional stages
 - Zombie worm *Osedax*
 - Degradation of organic matter (both flesh and intrusion of bones) releases sulfides.
- Wood falls are likely to happen along forested margins and near the mouths of rivers
 - Decay of cellulose and lignin
 - Wood-boring bivalve *Xylophaga*
- Other organic falls: kelp, alligator, jellyfish, turtle.

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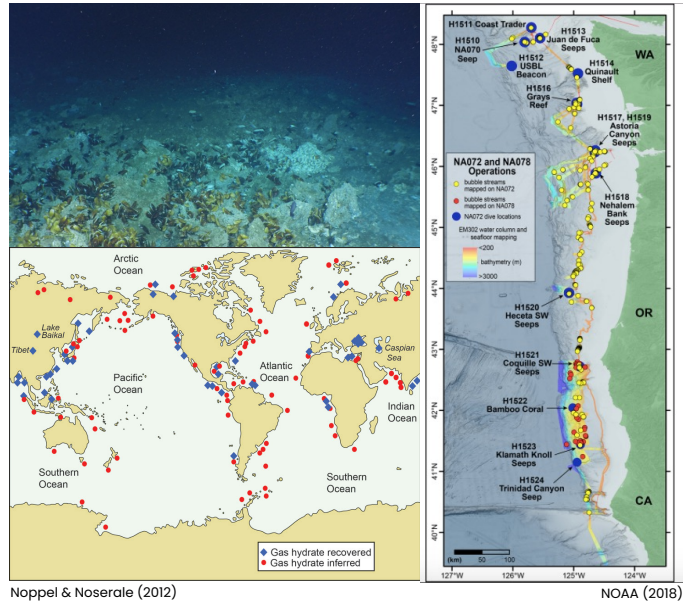
Oxygen Minimum Zones

- On productive continental margins (upwelling)
- Usually between 400-1,000 m deep
- Decomposition of organic matter uses up oxygen
- Limited fauna (nematodes)

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Methane seeps

- On continental margins associated with high tectonic activity (faults).
- Energy from the oxidation of sulfide, methane and hydrogen.
- Microbial mats, bathymodiolin mussels, vesicomyid clams, siboglinid polychaetes.
- Important role in biogeochemical cycling: Climate regulation and carbon sequestration



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Study questions

- How is a population different from a community? A niche from a habitat?
- What is a limiting factor and how is that related to range of tolerance of organisms? Give a few examples.
- How do you explain the richness and density of the intertidal zone? Relate your answer to intertidal zonation.
- What are the primary producers in kelp forests and coral reefs?
- List and explain 3 characteristics of wetlands.
- How are deep-sea ecosystems different than shallow-water ecosystems? Do they have anything in common?
- Pick 3 deep-sea ecosystems and compare them (food source, fauna, environmental conditions, etc).
- How are chemosynthetic ecosystems related? What is the stepping-stone hypothesis?
- What is deep-sea gigantism and why does it happen?
- List and explain 3 adaptations that deep-sea animals evolved mentioning what environmental factor they were overcoming.

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